Canonical Research Designs VI: Leniency Designs (aka Judge IV, aka Examiner IV)

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Roadmap for Today

- Today we're focusing on a particular research design: the *leniency design* (aka examiner/judge IV)

- These designs harness quasi-random assignment of decision-makers to estimate causal effects of high-stakes treatments

- Structure follows Goldsmith-Pinkham, Hull, and Kolesár (2025), "Leniency Designs: An Operator's Manual"

- Key topics:
  1. Setup and intuition
  2. Estimation: why UJIVE?
  3. Heterogeneous effects and monotonicity
  4. Inference
  5. A practical checklist

Part 1: Setup and Examples

High-level description of leniency designs

- In many applications, there is an administrator, judge, or examiner who plays an important role in deciding an outcome

- These outcomes include:
  - bail
  - bankruptcy
  - getting a loan
  - parole
  - disability insurance
  - patent granting
  - cancer screening

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- Key ingredients:
  1. as-good-as-random assignment of decision-maker
  2. discretion over a (typically binary) treatment
  3. heterogeneity in decision-maker behavior ("leniency")

Example 1: Bail setting in Philadelphia

- Stevenson (2018): Bail setting in Philadelphia
- After you're arrested, you can be:
  - held in jail
  - released on bail (you pay $)
  - released on recognizance (no bail $)
- This decision is made by one of six magistrates at preliminary hearings
  - The particular magistrate that a defendant faces depends on a rotating schedule
  - Hence, the randomness occurs from *the time of day*

Distortion of Justice: How the Inability to Pay Bail Affects Case Outcomes

Megan T. Stevenson*
George Mason University

This article uses a natural experiment to analyze whether incarceration during the pretrial period affects case outcomes. In Philadelphia, defendants randomly receive bail magistrates who differ widely in their propensity to set bail at affordable levels. Using magistrate leniency as an instrument, I find that pretrial detention leads to a 13% increase in the likelihood of being convicted, an effect largely explained by an increase in guilty pleas among defendants who otherwise would have been acquitted or had their charges dropped. I find also that pretrial detention leads to a 42% increase in the length of the incarceration sentence and a 41% increase in the amount of nonbail court fees owed. This latter finding contributes to a growing literature on fines-and-fees in criminal justice, and suggests that the use of money bail contributes to a "poverty-trap": those who are unable to pay bail wind up accruing more court debt. (*JEL* K14)

> I have had the "you can wait it out or take the deal and get out" conversation with way too many clients.
>
> —a public defender, Philadelphia

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Philadelphia employs six Arraignment Court Magistrates at a time, and one of the six will be on duty 24 hours a day, 7 days a week, including holidays. Each day is composed of three work shifts: graveyard (11:30 p.m.–7:30 a.m.), morning (7:30 a.m.–3:30 p.m.) and evening (3:30 p.m.–11:30 p.m.). Each magistrate will work for five days on a particular shift, take five days off, then do five days on the next shift, five days off, and so forth. For example, a magistrate may work the graveyard shift from January 1st to January 5th, have January 6th–10th off, then work the morning shift from January 11th–15th, have the 16th–20th off, do the evening shift from January 21st–25th, take the next five days off, and then start the cycle all over again.

Example 2: Bankruptcy judges

- Dobbie et al. (2017): Bankruptcy judges in Chapter 13
- When you file for Chapter 13 bankruptcy, you file a repayment plan to be approved by a judge
- This decision is made by different judges staffed at a court on a given day
  - The assigned judge is done randomly by the court
  - Hence, the randomness occurs within *location-time* of filing

Consumer Bankruptcy and Financial Health
Will Dobbie, Paul Goldsmith-Pinkham, and Crystal Yang
NBER Working Paper No. 21032
March 2015
JEL No. D14,K35

ABSTRACT

This paper estimates the effect of Chapter 13 bankruptcy protection on post-filing financial outcomes using a new dataset linking bankruptcy filings to credit bureau records. Our empirical strategy uses the leniency of randomly-assigned judges as an instrument for Chapter 13 protection. Over the first five post-filing years, we find that Chapter 13 protection decreases an index measuring adverse financial events such as civil judgments and repossessions by 0.316 standard deviations, increases the probability of being a homeowner by 13.2 percentage points, and increases credit scores by 14.9 points. Chapter 13 protection has little impact on open unsecured debt, but decreases the amount of debt in collections by $1,315.

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Bankruptcy judges are federal judges appointed to 14-year terms by the Court of Appeals in their judicial district. There are a total of 94 federal bankruptcy courts in the United States, including at least one bankruptcy court in each state, the District of Columbia, and Puerto Rico. Each bankruptcy court hears all cases originating from counties in its jurisdiction, and are often further divided into offices that hear all cases originating from a subset of counties in the court's jurisdiction. Bankruptcy judges often hear cases across multiple offices within their court, but only hear cases filed in their bankruptcy court. These cases are typically assigned to judges using a random number generator or a blind rotation system within each office.[8]

A simple motivating example: Two examiners

- Consider a stylized patent setting (Farre-Mensa et al. (2020)):
  - n patent applications, randomly assigned to one of two examiners s ("soft") or t ("tough")
  - Treatment $x_i \in \{0, 1\}$: patent granted or not
  - Outcome y_i: future patent applications by the firm

- Outcome equation (constant effects for now):

$$y_i = \gamma + \beta x_i + \varepsilon_i$$

- OLS is biased: firms with high ε_i (better scientists, more R&D) submit stronger applications $\Rightarrow$ positive selection bias

- Leniency design idea: use the randomly-assigned examiner's tendency to grant patents (their *leniency*) as an instrument for x_i

- With two examiners: using the examiner dummy z_i as an instrument is numerically identical to using estimated leniency $\hat{\ell}_i$

Part 2: Leniency, Many IVs, and Estimation

Notation: Many decision-makers and controls

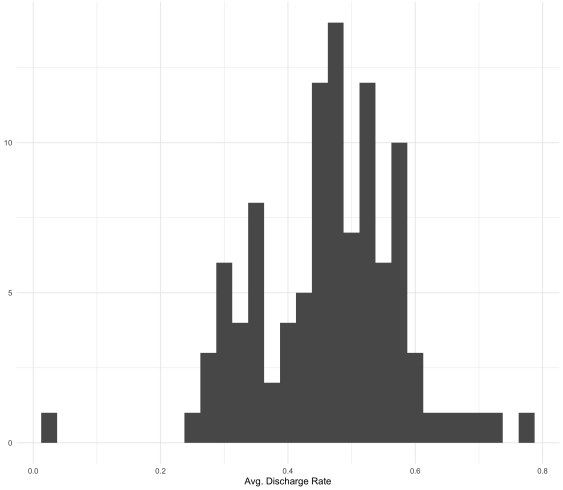
- Now consider K examiners. Assignment captured by instrument vector z_i (examiner dummies)
- **Necessary controls** w_i: variables needed to make assignment as-good-as-random
  - E.g. art unit $\times$ year FE (patents), court $\times$ time FE (judges)
- First-stage equation:

$$x_i = z_i'\pi + w_i'\delta + \nu_i$$

- Population first-stage fitted values give **absolute leniency**: $\ell_i = z_i'\pi + w_i'\delta$
- After residualizing for w_i, we get **relative leniency**: $\tilde{\ell}_i = \tilde{z}_i'\pi$
  - Measures leniency of assigned examiner *relative to* other examiners who could have been assigned
  - E.g. examiner's granting rate minus art unit average granting rate

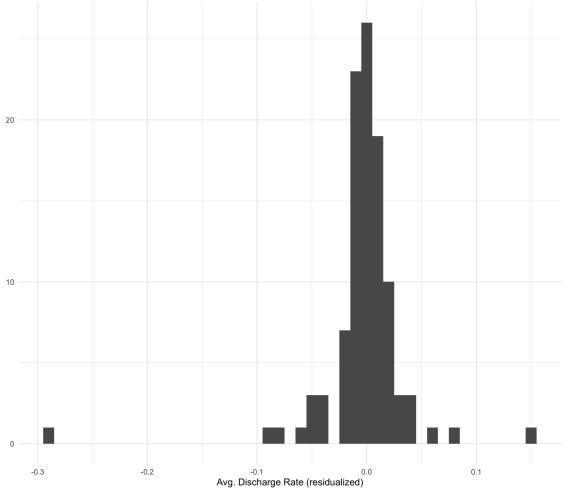
What's the instrument that people have used?

- In many papers, the measure that has been used is a "leniency" metric for judges: e.g. the $\hat{\ell}_i$
- This is the overall average leniency across judges in DGPY [note, not observation weighted – hence outliers]
- What's the problem with just the overall?
  - Judges are not random overall – location (and time) specific
- Must control for w_i to get *relative* leniency



Relative leniency after residualizing

- Once you residualize for controls, you get a much more recentered object
- This captures within-location variation and gives true examiner-specific variation
- Due to the location effect, we can't estimate "true" judge leniency – just relative leniency within an office



The infeasible ideal and the bias problem

- If we knew π, we could use relative leniency $\tilde{\ell}_i = \tilde{z}_i'\pi$ as the instrument:

$$\hat{\beta}^* = \frac{\sum_i y_i \tilde{\ell}_i}{\sum_i x_i \tilde{\ell}_i}$$

- Since $\tilde{\ell}_i$ is as-good-as-random and uncorrelated with ε_i, this is approximately unbiased

- But we don't know π! Using sample estimates $\hat{\pi}$ gives us 2SLS:
  - The estimated leniency $\hat{\ell}_i = \tilde{z}_i'\hat{\pi}$ is constructed *in part from observation i's own treatment*
  - This creates a mechanical correlation with $\varepsilon_i \Rightarrow$ many-IV bias toward OLS

How severe is the many-IV bias?

- Under homoskedasticity, 2SLS bias $\approx$ OLS bias $\times \frac{1}{(1-R^2)\times E[F]}$
  - $E[F]$: expected first-stage F statistic; R^2: partial R-squared from instruments
  - Since $R^2 \approx 0$ typically, the magnitude of $E[F]$ tells us how much smaller 2SLS bias is vs. OLS
- This justifies the $F > 10$ rule of thumb (Staiger and Stock, 1997): if $E[F] < 10$, 2SLS bias exceeds 10% of OLS bias
- With many examiners, F can be modest even when leniency explains meaningful variation, because the formula divides by K

The leave-out solution and why manual leniency is fragile

- The bias comes from using own treatment to estimate leniency

- Common practice: use a "leave-one-out" (LOO) leniency measure $\hat{\ell}_{-i}$
  - Excludes observation i's own treatment from the examiner average
  - Removes the mechanical correlation with ε_i

- But this "manual" leniency construction is fragile!
  - LOO of *absolute* leniency $\Rightarrow$ residualizing for w_i reintroduces own-observation bias through the control means
  - This is JIVE (Phillips and Hale, 1977); its bias can be comparable to 2SLS when there are many controls:

$$\text{JIVE bias} \approx \text{2SLS bias} \times \frac{-E[F] \times L}{(E[F]-1)K - L}$$

  where L = number of controls, K = number of instruments

- **Takeaway**: subtle variations in how the leniency measure is constructed and accounts for covariates can have large consequences

Use UJIVE!

- **UJIVE** (Unbiased Jackknife IV, Kolesár 2013): the purpose-built solution
  - Estimates *relative* leniency using leave-out: $\hat{\ell}_{-i} = \tilde{z}_i' \hat{\pi}_{-i}$
  - First residualize instruments for w_i, *then* do leave-out estimation
  - Avoids the own-observation bias of both 2SLS and JIVE

- Approximately unbiased even with many examiners and many controls

- Remains consistent even with weak instruments (as long as $\sqrt{K} \times E[F]$ is large)

- No need to report first-stage F as a diagnostic: UJIVE doesn't suffer from the bias that $F > 10$ screens for

- Don't construct leniency manually – use UJIVE as a one-step estimator
  - R package: `ManyIV` at `github.com/kolesarm/ManyIV`

Comparison of estimators

- **2SLS with examiner dummies**: biased toward OLS with many examiners
- **JIVE** (leave-out absolute leniency): bias reintroduced by many controls
- **IJIVE** (Ackerberg and Devereux, 2009): reverses order—residualize *then* leave-out; similar to UJIVE in practice
- **FEJIV** (Chao, Swanson, Woutersen, 2023): minimal MSE leniency with orthogonality to controls; attractive but computationally demanding
- **UJIVE**: recommended default
  - Minimal MSE without orthogonality constraint
  - Simple, established unbiasedness properties
  - Works with many instruments *and* many controls
- **Manual leniency IV**: widely advised against (like "manual 2SLS")

Part 3: Heterogeneous Effects and Monotonicity

What does UJIVE estimate with heterogeneous effects?

- So far we assumed constant β. In practice, treatment effects β_i vary across individuals
- The LATE theorem (Imbens and Angrist, 1994) provides a causal interpretation under:
  1. As-good-as-random assignment (conditional on controls)
  2. Exclusion restriction
  3. **Monotonicity**: if examiner *a* would grant in a case where *b* wouldn't, then *a* is more lenient in *all* cases
- With two examiners: divides population into **compliers** (treatment depends on assignment) and **non-responders** (always/never treated)
- Monotonicity rules out **defiers** (treated only by the tough examiner)
- IV estimates the average treatment effect for compliers (LATE)

Monotonicity with many examiners

- With many examiners, standard monotonicity is very strong: requires all examiners to rank *every* case identically
- Equivalent to assuming examiners differ only in their *cutoff*, not their ranking criteria
- Two core concerns:
  1. Multi-dimensional evaluation criteria (e.g. novelty vs. usefulness vs. non-obviousness)
  2. Skill variation induces ranking differences through mistakes

EXAMPLE 2 (Administrative Screening):[5] Suppose applicants for a social program are screened by two officials. The two officials are likely to have different admission rates, even if the stated admission criteria are identical. Since the identity of the official is probably immaterial to the response, it seems plausible that Condition 1 is satisfied. The instrument is binary so Condition 3 is trivially satisfied. However, Condition 2 requires that if official A accepts applicants with probability $P(0)$, and official B accepts people with probability $P(1) > P(0)$, official B must accept *any* applicant who would have been accepted by official A. This is unlikely to hold if admission is based on a number of criteria. Therefore, in this example we *cannot* use Theorem 1 to identify a local average treatment effect nonparametrically despite the presence of an instrument satisfying Condition 1.

Average monotonicity: a weaker condition

- With many examiners, UJIVE estimates a weighted average of pairwise-complier treatment effects, weighted by squared leniency differences
  - Larger complier groups get more weight
- Without monotonicity, defier groups get *negative* weight $\Rightarrow$ problematic
- But each individual appears in many pairwise comparisons. Total weight on individual i is their *average* complier/defier status
- **Average monotonicity** (Frandsen et al. 2023): the average leniency of examiners who would grant exceeds the average leniency of those who would deny
  - Weaker than standard monotonicity: allows some pairwise violations
  - Sufficient for positive individual weights in UJIVE
  - With two examiners, reduces to standard monotonicity

What if monotonicity is violated?

- Even with some negative weights, it only matters if negatively-weighted observations have systematically different treatment effects
- Sigstad (2025): in judicial panels where multiple judges see the same case, monotonicity is technically violated, but ranking disagreements are not severe enough to create substantial bias
- Key question: are the violations *large enough* to matter?

$$\beta^* = \frac{E[\tilde{\ell}_i y_i]}{E[\tilde{\ell}_i x_i]}.$$

The next result shows that β^* can be written as a weighted average of simple IV regressions restricted to the subpopulation with $w_i = w$ and $k(i) = k$ or $k(i) = j$. These simple IV regressions identify

$$\beta(j,k,w) = \frac{E[y_i \mid k(i) = k, w_i = w] - E[y_i \mid k(i) = j, w_i = w]}{\ell(k,w) - \ell(j,w)}.$$

Lemma 2. *The leniency IV estimand β^* may be written as a weighted average of pairwise IV regressions,*

$$\beta^* = \frac{\sum_{k>j} E[\omega(j,k,w_i)\beta(k,j,w_i)]}{\sum_{k>j} E[\omega(j,k,w_i)]}, \quad \omega(j,k,w) = p_j(w)p_k(w)(\ell(k,w) - \ell(j,w))^2. \tag{7}$$

Testing average monotonicity with UJIVE

- Key insight (building on Abadie 2002, Kitagawa 2015): UJIVE can **characterize compliers**

- For a binary variable v_i (determined before assignment), construct $\tilde{y}_i = v_i \times x_i$ and run UJIVE with $\tilde{y}_i$ as outcome
  - Under average monotonicity, UJIVE estimates a convex-weighted average of v_i for compliers
  - This estimate must lie in $[0, 1]$ since $v_i \in \{0, 1\}$
  - A negative estimate or one exceeding 1 $\Rightarrow$ evidence against average monotonicity

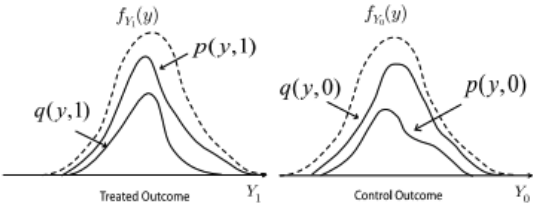
- Can also use the original outcome: set $\tilde{y}_i = y_i \times x_i$ to estimate weighted average of *treated* complier outcomes

- Create indicators for each outcome value (e.g. "zero future applications") and check all UJIVE estimates are positive

- This test inherits UJIVE's unbiasedness with many examiners/controls

Kitagawa (2015) result (background)

- Consider binary endogenous treatment D and a discrete instrument Z.
- Joint *testable* assumption: instrument is valid and monotonicity. Why?
  - Consider
    $p(y,d) = Pr(Y = y, D = d | Z = 1)$,
    $q(y,d) = Pr(Y = y, D = d | Z = 0)$
  - Let P and B be the probability over sets
  - Imbens and Rubin (1997) show
    $P(B,1) - Q(B,1) = Pr(Y_1 \in B, D_1 > D_0)$
    $P(B,0) - Q(B,0) = Pr(Y_0 \in B, D_1 > D_0)$



- Testable implication:

$$P(B,1) - Q(B,1) \geq 0$$
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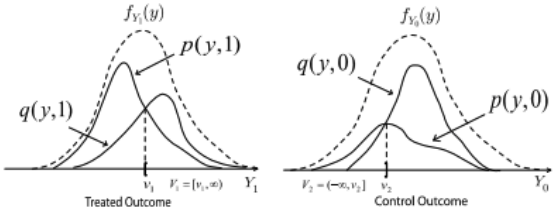

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Part 4: Inference

Standard errors: cluster or not?

- The assignment mechanism should guide the choice of standard errors
- **Individual-level randomization** (e.g. patent examiners assigned idiosyncratically within art unit-year):
  - Heteroskedasticity-robust (HC) standard errors are appropriate
  - No need to cluster by examiner
- **Group-level assignment** (e.g. all cases in a month assigned to the same examiner):
  - Cluster at the group level (e.g. month)
  - May also need leave-cluster-out modification of UJIVE (Frandsen et al. 2025)
- Rule of thumb: clustering is needed when groups of observations are assigned together
- UJIVE computes correct heteroskedastic-robust SEs automatically
- Clustering justification also has implications for the *estimator*, not just inference

Don't be fooled by small 2SLS standard errors

- With many examiners, 2SLS standard errors can be *much* smaller than UJIVE SEs
- This reflects statistical pathology, not efficiency:
  - 2SLS estimates are pulled toward OLS (biased)
  - 2SLS standard errors are also pulled toward OLS SEs (too small)
  - Creates a false sense of precision around biased estimates
- Even leave-out leniency measures (manual construction) can produce SEs that don't reflect estimation error in the leniency measure
- UJIVE delivers both unbiased point estimates *and* standard errors that accurately capture sampling variation

Part 5: A Practical Checklist

The 5-step checklist (GHK 2025)

1. **Identify necessary controls** for as-good-as-random assignment, and determine appropriate estimator and standard errors
2. **Verify balance** on predetermined observables
3. **Estimate treatment effects** by UJIVE (and compare to 2SLS/OLS)
4. **Test average monotonicity** to assess LATE interpretation
5. **Characterize compliers** to assess external validity

Step 1: Identify necessary controls and design

- Begin with institutional knowledge that justifies quasi-random assignment
  - What is the assignment mechanism? What must be conditioned on?
  - Patent example: assignment within art units is effectively random, conditional on the set of examiners $\Rightarrow$ art unit $\times$ year FE
  - Bail example: rotating schedule $\Rightarrow$ time-of-day variation
- These are *necessary controls*—must be included in every specification
- Separately, may include *precision controls* (pre-assignment covariates) to reduce SEs
- Also argue for the exclusion restriction from institutional knowledge
  - Patent examiners make narrow approval decisions $\Rightarrow$ unlikely to directly affect future innovation
- Assignment mechanism determines clustering level for SEs (and potentially for leave-cluster-out UJIVE)

Step 2: Verify balance on observables

- As-good-as-random assignment implies average predetermined characteristics are equal across examiners (conditional on controls)

- **Best practice**: run UJIVE specification with each predetermined covariate as the outcome
  - Same instruments, same controls as main specification
  - Avoids finite-sample biases from regressing on constructed leniency

- Magnitude of imbalance directly translates to potential bias in treatment effects

Table 2: Balance Checks

	Coefficient (1)	Std. Error (2)	# Obs. (3)
Patent class group I	-0.040	0.017	32,514
Patent class group II	0.011	0.021	32,514
Patent class group III	0.032	0.021	32,514
Log(# independent claims in application)	0.066	0.084	27,226
Log(1 + # VC rounds before application)	-0.024	0.035	32,514
Log(# startups in HQ state)	-0.273	0.143	32,514
Approval by European Patent Office	0.013	0.212	3,287
Approval by Japanese Patent Office	0.525	1.258	1,206

Notes: This table reports results of UJIVE regressions of each covariate in Panel C of Table 1 on application approval, instrumenting with examiner indicators. All estimates control for art unit-by-year fixed effects. The reported standard errors are robust to heteroskedasticity and treatment effect heterogeneity.

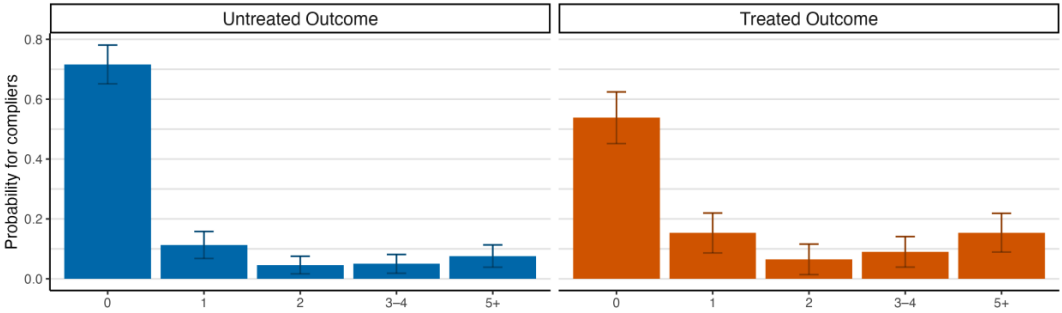
Step 3: Estimate treatment effects

- Use UJIVE as the preferred estimator
- Report alongside alternative estimators for comparison:
  - OLS (benchmark for selection bias)
  - 2SLS with examiner dummies (shows many-IV bias)
  - Manual leniency IV if used in prior literature (shows construction sensitivity)
- Pattern to look for:
  - 2SLS estimates between OLS and UJIVE $\Rightarrow$ 2SLS pulled toward OLS as predicted
  - 2SLS SEs much smaller than UJIVE SEs $\Rightarrow$ false precision from bias
- In Farre-Mensa et al. reanalysis: UJIVE finds significant positive effects of patent approval on future innovation, but 2SLS SEs are 3–4x smaller (pathological)

Step 4: Test average monotonicity

- For each possible outcome value, create indicator and interact with treatment:
  - E.g. $\tilde{y}_i = \mathbf{1}(y_i = k) \times x_i$ for each k
  - Run UJIVE with same instruments and controls
  - All estimates should be positive (and between 0 and 1)
- This tests the weaker *average* monotonicity, not full IA94 monotonicity
- Uses the same UJIVE framework—inherits robustness to many instruments/controls
- Frandsen et al. (2023) test of strict monotonicity also available (`testjfe` in Stata), but:
  - Rules out many decision-makers
  - Requires bounded outcomes
  - Tests the stronger condition

(a) # of subsequent applications



Step 5: Characterize compliers for external validity

- Even with a valid LATE, we want to know: are compliers representative of the broader population?

- Use UJIVE with modified outcomes to characterize compliers:
  - For each baseline covariate v_i, run UJIVE with $\tilde{y}_i = v_i \times x_i$ as outcome
  - Recovers weighted average of v_i for treated compliers
  - For untreated compliers: use $v_i \times (1 - x_i)$ instead
  - Pool efficiently: use $\tilde{x}_i = 2x_i - 1$ and regress $v_i \times \tilde{x}_i$ on $\tilde{x}_i$ (Angrist et al. 2023)

- Compare complier means to population means
  - If similar: LATE likely approximates ATE
  - If different: effects may not generalize

- Complier estimates within $[0, 1]$ for binary v_i also supports average monotonicity

Summary: Key recommendations

- **Estimation**: Use UJIVE, not manual leniency or 2SLS
  - Avoids many-IV bias and many-control bias simultaneously
  - R package: `ManyIV`

- **Controls**: Identify necessary controls from institutional knowledge; precision controls optional

- **Inference**: Heteroskedastic-robust SEs when individual-level assignment; cluster when group assignment

- **Monotonicity**: Test average monotonicity (weaker than standard) via UJIVE

- **External validity**: Characterize compliers using UJIVE with modified outcomes

- **Don't** construct leniency measures manually – one-step UJIVE is simpler and safer

Further Readings

- Goldsmith-Pinkham, Hull, and Kolesár (2025), “Leniency Designs: An Operator’s Manual”
- Frandsen, Lefgren, and Leslie (2023), “Judging Judge Fixed Effects”
- Kolesár (2013), “Estimation in an Instrumental Variables Model with Treatment Effect Heterogeneity”
- Cunningham (2021) chapter 7.8.2
- Farre-Mensa, Hegde, and Ljungqvist (2020)
- Stevenson (2018); Dobbie, Goldsmith-Pinkham, and Yang (2017)